



Scenario Description

CodeWorks Inc. develops software for external clients. As an input, they receive requirements, but their quality varies significantly. The developers of the company have long suspected that the quality of these input requirements affect the volume of communication around them and the number of misunderstandings they cause. Now, they are looking for a reliable answer to these questions.

But the issue is not that simple. Some developers report that requirements quality does not affect communication volume directly, but rather causes frustration, and that frustration - in turn - affects the communication volume. Others hypothesize that both the requirements quality, but also the communication volume and the number of misunderstandings is caused by the complexity of the software they are currently working on. In CodeWorks' requirements backlog, requirements are escalated as critical if either their requirements quality is perceived to be too low or the number of misunderstandings it amasses is too high.

Without disrupting operations, CodeWorks Inc. observed their active working environment to answer the two questions:

1. "Does the quality of the input requirements affect the communication volume?"
2. "Does the quality of the input requirements affect the number of misunderstandings?"

Thanks to the observation, the following data is available to you:

Column	Description	Type
id	Unique identifier of the observation	ID
complexity	Complexity of the software	Numeric
req_quality	Quality of the current requirement	Numeric
frustration	Measure of the perceived frustration	Numeric
comm_volume	Volume of communication	Numeric
misunderstandings	Number of reported misunderstandings	Count
escalated	Whether the requirement was flagged as critical	Binary